

STARspatial Topography Accessibility Robot

Locked Inc. | Texas A&M University | ESET Senior Capstone, Spring 2026 | github.com/Locked-Inc/STAR

Elevator pitch

STAR is a distributed autonomous ground-robotics platform that provides the measurement layer for continuous indoor ADA compliance auditing. A Raspberry Pi 5 runs ROS2 Jazzy (SLAM, EKF, Nav2, frontier exploration). A custom Renesas RX72N PCB handles 250 Hz real-time motor control. The two sides talk over a 10 Mbps SPI link with HARQ, FEC, and nanpb Protocol Buffers. On top of the platform, we have built an ADA compliance engine that implements the ramp-slope check end-to-end and architects six more against the same sensor stack.

Why this matters

More than 1 in 4 U.S. adults (~70 million) live with a disability (CDC 2022 BRFSS, released July 2024). Industry estimates put ~73% of U.S. commercial buildings out of compliance; plaintiffs filed **8,800 ADA Title III federal lawsuits in 2024** (+7% YoY, Seyfarth Shaw). DOJ civil penalties are **\$118,225 first / \$236,451 repeat** per 90 FR 29445 (July 2025). Only 32% of commercial property buyers even ask about ADA during inspection (Focus Building Inspections).

Sensor stack (all physically wired)

Sensor	Role
RPLiDAR C1 (360, 10 Hz, 12 m)	Primary geometry, SLAM, ramp plane, path width
Waveshare IMX219-83 stereo	Dual 8 MP Sony IMX219, 60 mm baseline, dual MIPI CSI-2
BNO055 9-DoF IMU (chassis)	Ramp-slope pitch cross-validation, jolt detection
ICM20948 9-DoF IMU (camera)	Visual-inertial reference for stereo pipeline
4x HC-SR04 ultrasonic	Short-range obstacle safety
DS18B20 + BMP280	Temperature + barometric telemetry

Compute and transport

Raspberry Pi 5 runs ROS2 Jazzy (star_bringup, star_spi_bridge, star_gateway_bridge, star_safety_monitor), the Go gateway (gRPC-Web), and the React UI. **RX72N** (R5F572NNHxFB, 4 MB Flash, 1 MB SRAM, ThreadX RTOS) runs 250 Hz PID motor control. **SPI 10 Mbps** with SYNC-0x55AA framing, CRC-32, HARQ Chase Combining, convolutional FEC K=7 rate-1/2, and nanpb protobuf. Peak utilization 1.6%.

Compliance engine - implementation status

Check	ADA	Status
Ramp slope > 1:12	405.2	IMPLEMENTED
Trip hazard > 0.25 in	303	STRETCH
Path width < 36 in	403.5	STRETCH
Ramp width < 36 in	405.5	architected
Ramp landing < 60x60 in	405.7	architected
Door clear width < 32 in	404.2.3	architected
Door threshold > 0.5 in	404.2.5	architected

Validation

Platform: 143/143 ROS2 tests passing across 4 packages. slam_toolbox async emits /map at 0.5 Hz on RPi5 with full TF chain. Nav2 autonomous goals verified; m-explore-ros2 frontier exploration to completion. 250 Hz PID tuned against first-order motor model (tau = 75 ms).

Compliance (ramp slope): ground truth via Wixey WR300 digital angle gauge (+/- 0.1 deg). Cross-validation gate between BNO055 pitch and LiDAR plane normal at +/- 0.5 deg. Validation log in extras/STAR_ValidationLog.xlsx. Numbers on poster and deck come from the log, not estimates.

Impact at TAMU scale

Texas A&M's College Station campus covers 5,200 acres with hundreds of buildings. Any manual CASp audit pass is a six- or seven-figure cost and runs months of specialist time. STAR re-uses one robot across buildings at amortized hardware cost - the same economic shift cybersecurity scanners brought to IT compliance. Stakeholders under discussion: TAMU Department of Disability Resources (user advocacy), Office of Risk, Ethics & Compliance (compliance program), Facilities Services (remediation). Code is open source under the Locked-Inc GitHub organization.

Team Locked Inc.

Four-member ESET capstone team. Contact on reverse.

Sources (verified against primary documents; full matrix at [research/source_verification.md](#) and BibTeX at [extras/references.bib](#)): Seyfarth Shaw ADA Title III Report March 2025; Federal Register 90 FR 29445 (2025-12494); CDC DHDS 2022 BRFSS released July 2024; Okoro et al. MMWR 2018; 67(32):882-887; Sun et al. Sensors 2024, 24(21):6828; ADA 2010 Standards for Accessible Design.